

UNIT-III

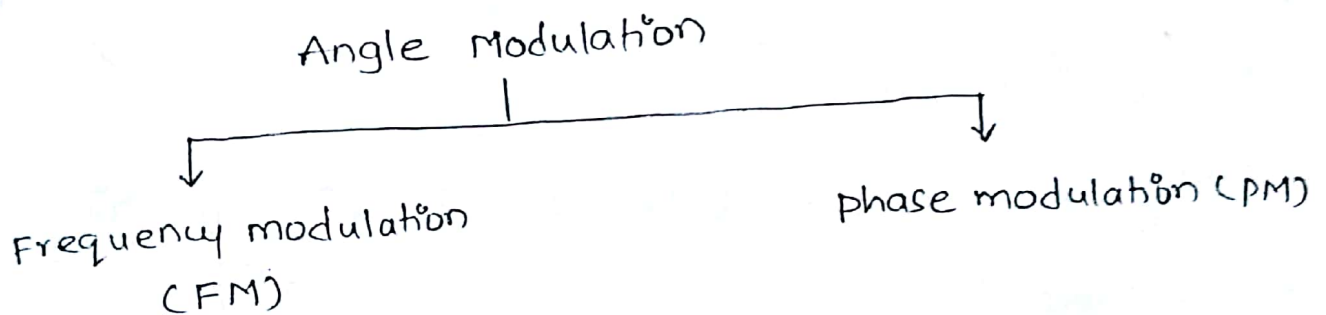
ANGLE MODULATION

Syllabus: Introduction, Basic concept of phase modulation, Frequency Modulation: Single tone frequency modulation, Spectrum Analysis of Sinusoidal FM Wave, Narrow band FM, Wide band FM, Constant Average Power, Transmission bandwidth of FM Wave, Generation of FM Waves: Direct Method, Indirect Method, Detection of FM Waves: Balanced Frequency discriminator, Zero crossing detector, Phase locked loop, Comparison of FM & AM, Related problems.

Angle Modulation concepts

Introduction:-

Angle modulation may be defined as the process in which the total phase angle of a carrier wave is varied in accordance with the instantaneous amplitude of the message signal.

Mathematical Representation:-

let us consider that an unmodulated carrier signal is expressed as

$$c(t) = A \cos(\omega_c t + \phi)$$

where A = Amplitude of carrier

ω_c = carrier frequency

ϕ = some phase angle.

Substituting $\omega_c t + \phi = \theta(t)$ we get

$$c(t) = A \cos[\theta(t)]$$

$\theta(t) = \omega_c t + \phi$ is the total phase angle of the carrier

Let 'c' be a continuously rotating phasor, given

by
$$c = A e^{j\theta} = A e^{j\theta(t)}$$

then
$$c(t) = \text{Re}(A e^{j\theta}) = A \cos(\theta(t))$$

Let the phasor 'c' rotates at a constant angular velocity ω_c and ϕ is the phase angle of the unmodulated carrier at $t=0$.

The constant angular velocity ω_c is related to total phase angle θ as from eq (1)

$$\theta = \omega_c t + \phi$$

Differentiating both sides of this equation with respect to t we have

$$\omega_c = \frac{d\theta(t)}{dt}$$

The derivative $\frac{d\theta(t)}{dt}$ is constant for unmodulated carrier, However this derivative $\frac{d\theta(t)}{dt}$ may not be a constant with time. In general this may vary with time.

Therefore the angular velocity of the phasor 'c' would also vary with time.

Hence this time dependent angular velocity or angular frequency is known as "Instantaneous angular velocity" or "instantaneous angular frequency". This instantaneous angular frequency is denoted by ω_i then

$$\omega_i = \frac{d\theta_i(t)}{dt}$$

$$2\pi f_i = \frac{d\theta_i(t)}{dt}$$

$$f_i = \frac{1}{2\pi} \frac{d\theta_i(t)}{dt}$$

Integrating

$$\theta_i = \int \omega_i dt$$

Types of angle modulation:-

- (i) phase modulation (PM)
- (ii) Frequency modulation (FM)

i) phase modulation (PM):-

In phase modulation the phase of the carrier signal changes in accordance with the instantaneous amplitude of the message signal by keeping amplitude and frequency of carrier signal constant.

Here the instantaneous phase of carrier signal is proportional to the message signal

$$\theta_i(t) \propto m(t)$$

it can be related as

$$\theta_i(t) = 2\pi f_c t + k_p m(t)$$

Let the unmodulated carrier as

$$c(t) = A_c \cos[2\pi f_c t + \phi]$$

In general an angle modulated wave can be written as

$$s(t) = A_c \cos \theta_i(t)$$

But in phase modulation

$$\theta_i(t) = 2\pi f_c t + k_p m(t)$$

$\therefore$ The time domain equation of phase modulation is

$$s(t) = A_c \cos[2\pi f_c t + k_p m(t)]$$

By comparing ^{modulated} wave with the unmodulated carrier we may write as

$$\boxed{\phi = K_p m(t)} \text{ rad.}$$

K_p = phase sensitivity of modulator (rad/volt).

Here voltage variations of signal are converted into phase variations.

For single tone modulation:-

• let $m(t) = A_m \cos 2\pi f_m t$

$$s(t) = A_c \cos [2\pi f_c t + K_p A_m \cos 2\pi f_m t]$$

let $\boxed{K_p A_m = \Delta\phi}$

where $\Delta\phi$ = phase deviation.

• $s(t) = A_c \cos [2\pi f_c t + \Delta\phi \cos 2\pi f_m t]$

$$\boxed{s(t) = A_c \cos [2\pi f_c t + \beta \cos 2\pi f_m t]}$$

where β = modulation index

∴ The phase deviation itself taken as modulation index here.

Frequency Modulation:-

Frequency modulation is defined as the process in which the frequency of the carrier signal is varied in accordance with the instantaneous amplitude of message signal by keeping Amplitude and phase of the carrier signal is constant.

In FM the instantaneous angle of carrier signal is proportional to the integral of message signal

$$\theta_c(t) \propto \int m(t) dt$$

In FM the instantaneous frequency of the carrier is related with $m(t)$ as

$$f_i = f_c + K_f m(t)$$

where K_f = frequency sensitivity of the modulator.

$$K_f = \text{Hz/volts.}$$

K_f indicates the change in carrier frequency for 1V of message signal.

In general an angle modulated wave can be written as

$$s(t) = A_c \cos \theta_i(t)$$

But we know that

$$f_i = \frac{1}{2\pi} \frac{d\theta_i(t)}{dt}$$

$$\theta_i(t) = 2\pi \int f_i dt$$

$$\text{But } f_i = f_c + k_f m(t)$$

$$\theta_i(t) = 2\pi \int [f_c + k_f m(t)] dt$$

$$\theta_i(t) = 2\pi f_c t + 2\pi k_f \int m(t) dt$$

The time domain equation of FM wave is

$$s(t) = A_c \cos[\theta_i(t)]$$

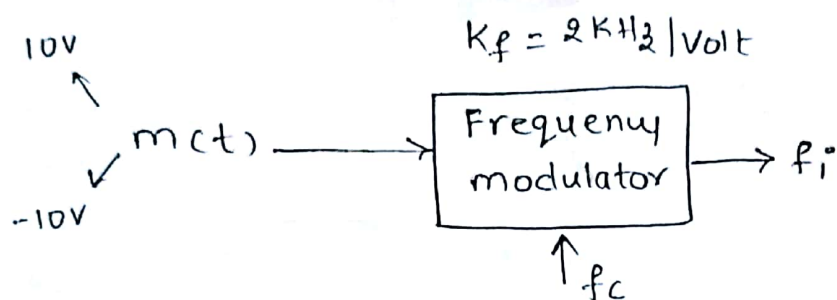
$$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t) dt]$$

Frequency deviation:-

We know that the instantaneous frequency of FM wave is given as

$$f_i = f_c + k_f m(t)$$

k_f indicates the change in carrier frequency for 1V of message signal.



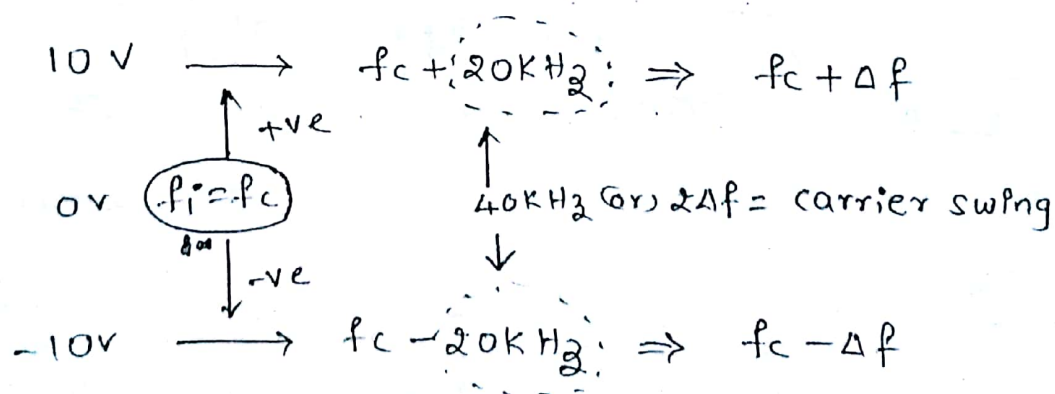
$$m(t) = 0 \Rightarrow f_i = f_c$$

$$m(t) = 2\text{V} \Rightarrow f_i = f_c + 4\text{kHz}$$

$$m(t) = 5\text{V} \Rightarrow f_i = f_c + 10\text{kHz}$$

$$m(t) = -10\text{V} \Rightarrow f_i = f_c - 20\text{kHz}$$

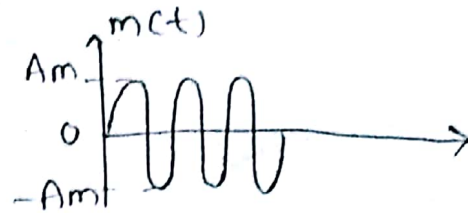
$\therefore$ FM modulator consider as voltage to frequency converter.



where Δf = frequency deviation.

∴ The carrier swing = $2\Delta f$

let $m(t) = A_m \cos 2\pi f_m t$



$$f_{\max} = f_c + K_f A_m = f_c + \Delta f$$

$$f_{\min} = f_c - K_f A_m = f_c - \Delta f$$

where $\Delta f = K_f A_m$

The instantaneous frequency of FM signal varies with time around the carrier frequency f_c . This means that the instantaneous frequency of FM signal varies according to the message signal.

* The maximum change in instantaneous frequency from the average frequency f_c is called "frequency deviation".

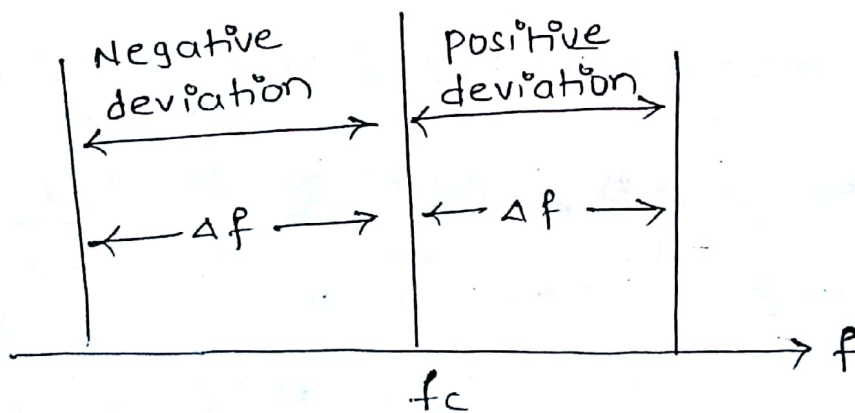


fig: illustration of frequency deviation.

Single tone modulation:-

$$\text{let } m(t) = A_m \cos 2\pi f_m t$$

The FM wave is

$$s(t) = A_c \cos[2\pi f_c t + 2\pi K_f \int m(t) dt]$$

$$s(t) = A_c \cos[2\pi f_c t + 2\pi K_f \int A_m \cos 2\pi f_m t dt]$$

$$s(t) = A_c \cos[2\pi f_c t + \frac{2\pi K_f A_m}{2\pi f_m} \sin 2\pi f_m t]$$

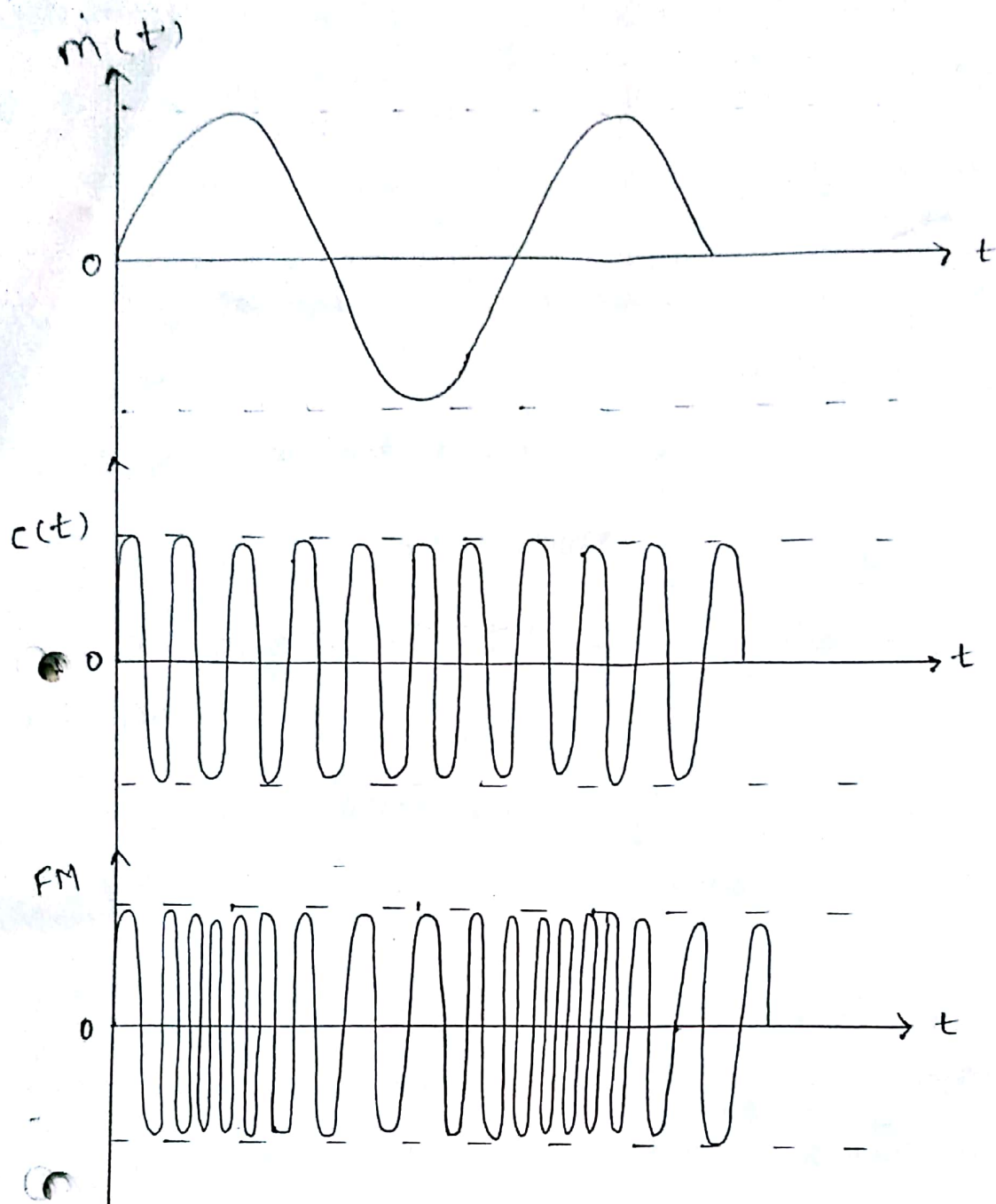
$$s(t) = A_c \cos[2\pi f_c t + \left(\frac{K_f A_m}{f_m}\right) \sin 2\pi f_m t]$$

$$\text{let } \beta = \frac{K_f A_m}{f_m}$$

$$\therefore \boxed{s(t) = A_c \cos[2\pi f_c t + \beta \sin 2\pi f_m t]}$$

where β = modulation index.

$$\beta = \frac{K_f A_m}{f_m} = \frac{\Delta f}{f_m} = \frac{\text{frequency deviation}}{\text{message frequency}}$$



Few points about FM :-

- 1) The instantaneous carrier frequency f_i deviates up and down from its centre frequency f_c . This change or shift above or below the carrier frequency is called frequency deviation denoted by Δf .

$$\Delta f = k_f m(t) |_{\max.}$$

- 1) The total variation in frequency from the lowest to the highest point is called "carrier swing".

$$\text{The carrier swing} = 2 \times \text{frequency deviation} = 2\Delta f.$$

- 3) The amount of frequency deviation depends upon the amplitude (loudness) of the modulating (audio) signal. This means that louder the sound, greater the frequency deviation.

For FM broadcasting $\Delta f = 75 \text{ KHz}$ on each side of f_c

For sound portion of TV broadcasting

$$\Delta f = 25 \text{ KHz}.$$

- 4) Frequency deviation is useful in determining the FM signal BW, since a maximum frequency deviation of 75 KHz is allowed for commercial FM broadcast stations using a band of 88 MHz to 108 MHz , therefore approximate FM channel width is $2 \times 75 = 150 \text{ kHz}$. Allowing a 25 KHz guard band on either side, the channel width becomes $2(75 + 25) = 200 \text{ kHz}$. This guardband is meant to prevent interference between adjacent channels.

** In FM broadcasting, the highest audio frequency transmitted is 15 KHz .

5) For FM, the modulation index is defined as the ratio of frequency deviation to the modulating frequency.

$$\beta = \frac{\text{frequency deviation}}{\text{modulating frequency}} = \frac{\Delta f}{f_m}$$

This modulation index may be greater than unity.

Relationship between phase modulation (PM) and frequency modulation (FM) :-

The expression for PM is

$$s(t) = A_c \cos[\omega_c t + k_p m(t)] \quad \text{--- (1)}$$

The expression for FM wave is

$$s(t) = A_c \cos[\omega_c t + 2\pi K_f \int_0^t m(t) dt] \quad \text{--- (2)}$$

From (1) & (2), it may be observed that PM & FM are closely related to each other because in both the cases there is a variation in the total phase angle.

In PM, phase angle varies linearly with baseband signal $m(t)$ whereas in FM, the phase angle varies linearly with the integral of baseband

signal $m(t)$). This means that FM wave may be obtained by using PM and viceversa.

To get FM by using PM, we first integrate the baseband signal and then apply to the phase modulator as shown below.

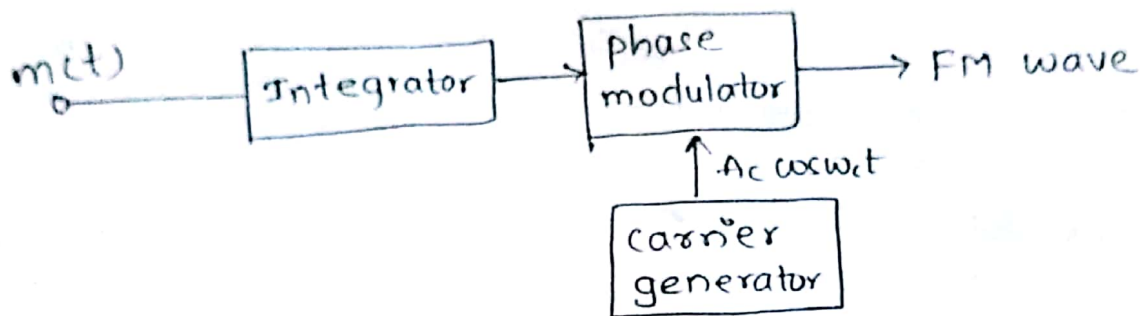


fig: Generation of FM using PM.

Illy PM wave may be generated by using frequency modulator by first differentiating the baseband signal $m(t)$ and then applying to the frequency modulator as shown below.

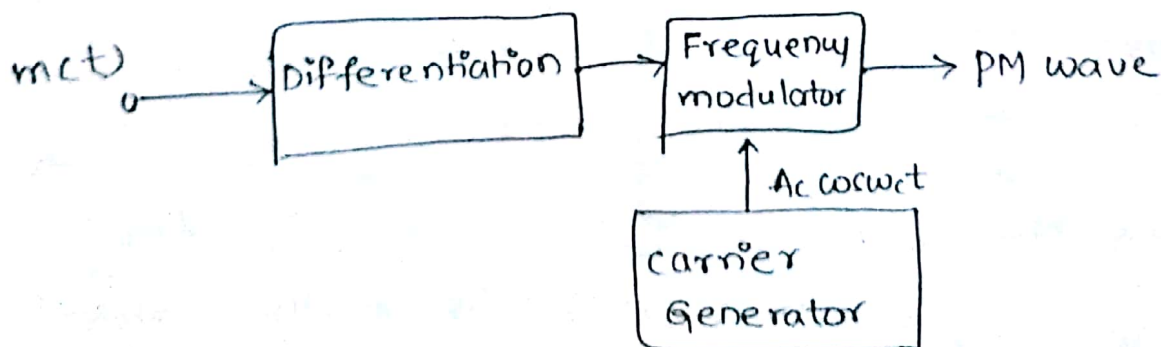
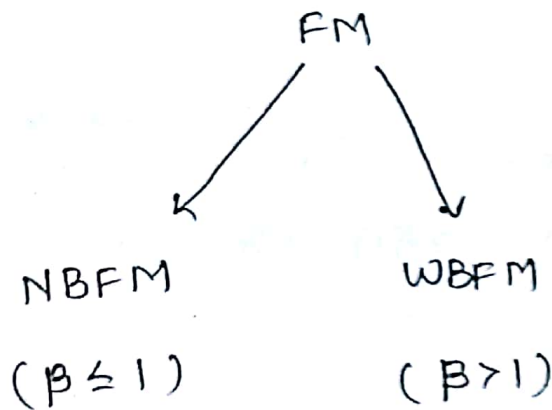


fig: Generation of PM using FM.

Types of Frequency modulation (FM):-

If the frequency deviation is high, the BW will be large and if the frequency deviation is low the BW will be small. Since deviation is given by $\Delta f = K_f m(t)$, it means that for given $m(t)$ the frequency deviation and bandwidth will depend upon frequency sensitivity K_f . Similarly the modulation index β also depends on K_f .

$\therefore$ Depending on the value of modulation index β we may distinguish two cases of frequency modulation 1) Narrow-band FM for which β is small and 2) wide band FM for which β is large. Both are compared to one radian.



Narrow-band frequency modulation (NBFM):-

For Narrow-band frequency modulation both modulation index and frequency deviation are very very small.

Let us find the NBFM expression for single tone modulation.

$$\text{Let } m(t) = A_m \cos 2\pi f_m t$$

The expression for FM wave is

$$s(t) = A_c \cos [2\pi f_c t + 2\pi K_f \int m(t) dt]$$

By substituting $m(t)$ in above equation we get

$s(t)$ as

$$s(t) = A_c \cos \left[\frac{2\pi f_c t}{A} + \frac{\beta \sin 2\pi f_m t}{B} \right]$$

$$- [\cos(A+B) = \cos A \cos B - \sin A \sin B]$$

$$s(t) = A_c \cos 2\pi f_c t \cos [\beta \sin 2\pi f_m t]$$

$$- A_c \sin 2\pi f_c t \sin [\beta \sin 2\pi f_m t]$$

for NBFM, $\beta \ll 1$

for smaller values of θ i.e. $\theta \ll 1$, $\sin \theta \approx \theta$
 $\cos \theta \approx 1$.

assume $\theta = \beta \sin 2\pi f_m t$

As $\beta \ll 1$ then $\theta \ll 1$

$$s(t) = A_c \cos 2\pi f_c t \cos[\theta] - A_c \sin 2\pi f_c t \sin[\theta]$$

$$\cos \theta \cong 1, \sin \theta \cong \theta$$

$$\therefore s(t) \cong A_c \cos 2\pi f_c t \cdot 1 - A_c \sin 2\pi f_c t \cdot \theta$$

$\therefore$ The time domain expression for NBFM is

$$s(t) \cong A_c \cos 2\pi f_c t - A_c \beta \sin 2\pi f_m t \sin 2\pi f_c t \quad \text{--- (1)}$$

(let $m(t) = A_m \sin 2\pi f_m t$ then

$$s(t) \cong A_c \cos 2\pi f_c t + A_c \beta \cos 2\pi f_m t \sin 2\pi f_c t \quad \text{--- (2)}$$

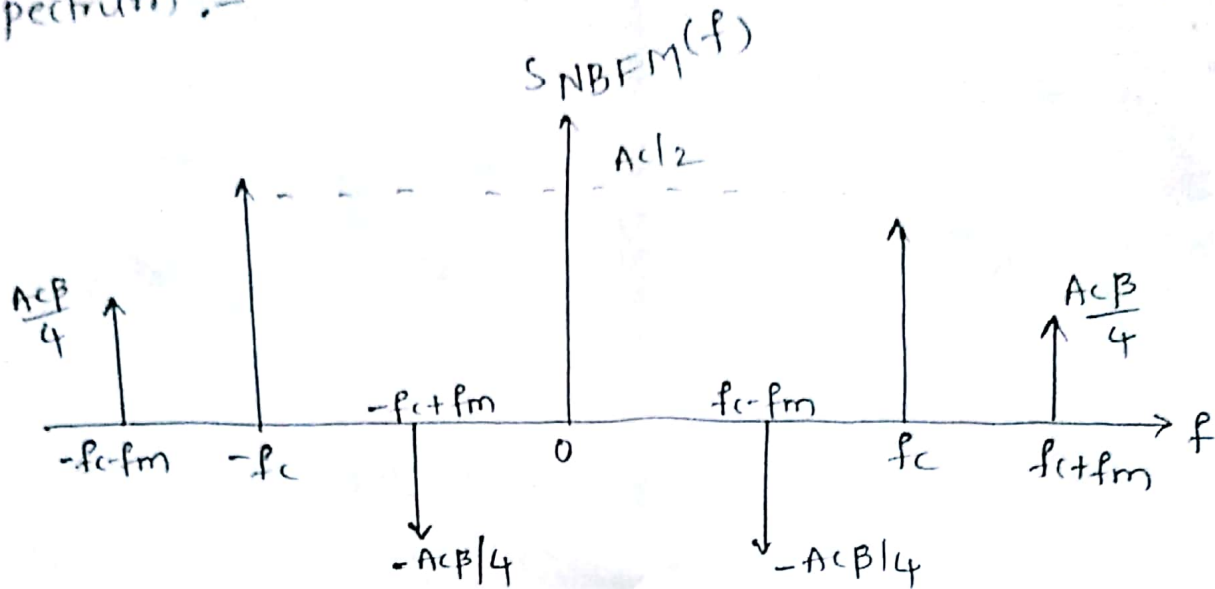
from equation (1)

$$s(t) \cong A_c \cos 2\pi f_c t - \frac{A_c \beta}{2} [\cos 2\pi (f_c - f_m) t - \cos 2\pi (f_c + f_m) t]$$

$$s(t) \underset{\text{NBFM}}{\cong} A_c \cos 2\pi f_c t - \frac{A_c \beta}{2} \cos 2\pi (f_c - f_m) t + \frac{A_c \beta}{2} \cos 2\pi (f_c + f_m) t$$

$\downarrow$ carrier $\downarrow$ LSB $\downarrow$ USB

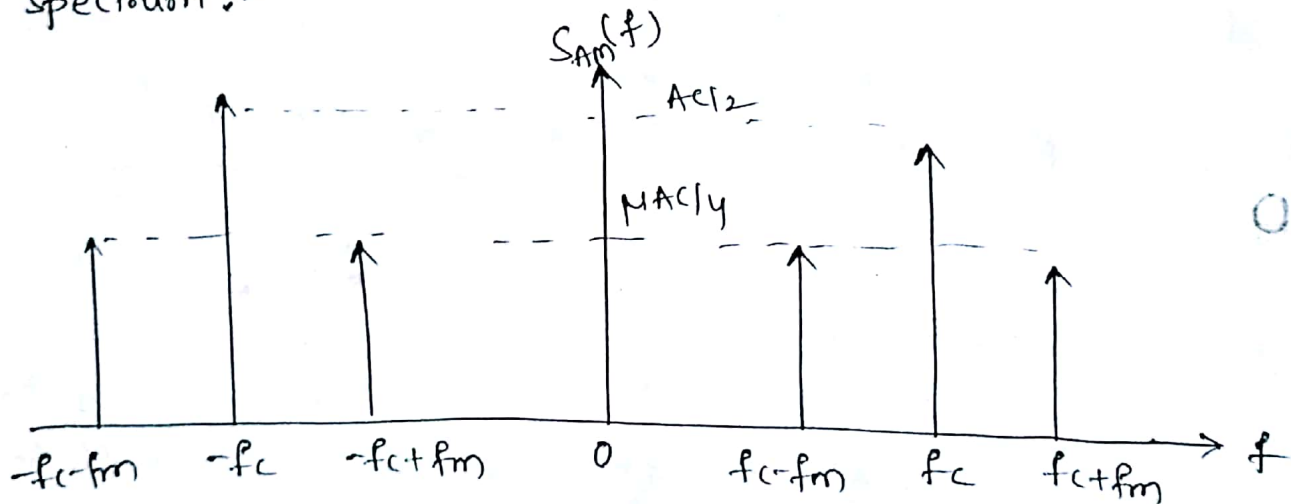
Spectrum :-



The NBFM wave expression is similar to AM wave expression.

$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{M A_c}{2} \cos 2\pi (f_c - f_m) t + \frac{M A_c}{2} \cos 2\pi (f_c + f_m) t$$

Spectrum :-



→ By comparing NBFM with AM we can say that, in NBFM LSB is out of phase by 180° .

* The transmission bandwidth of a narrow band FM wave is closely equal to twice the message bandwidth.

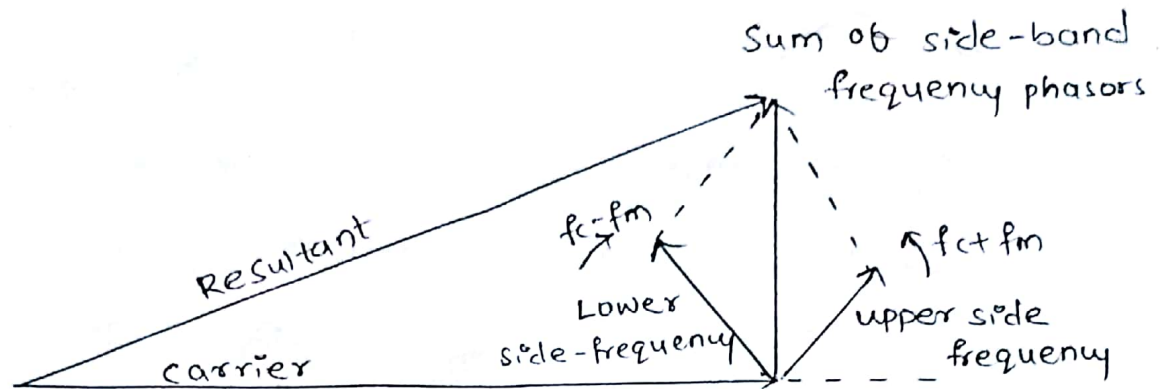


fig (a). phasor of NBFM.

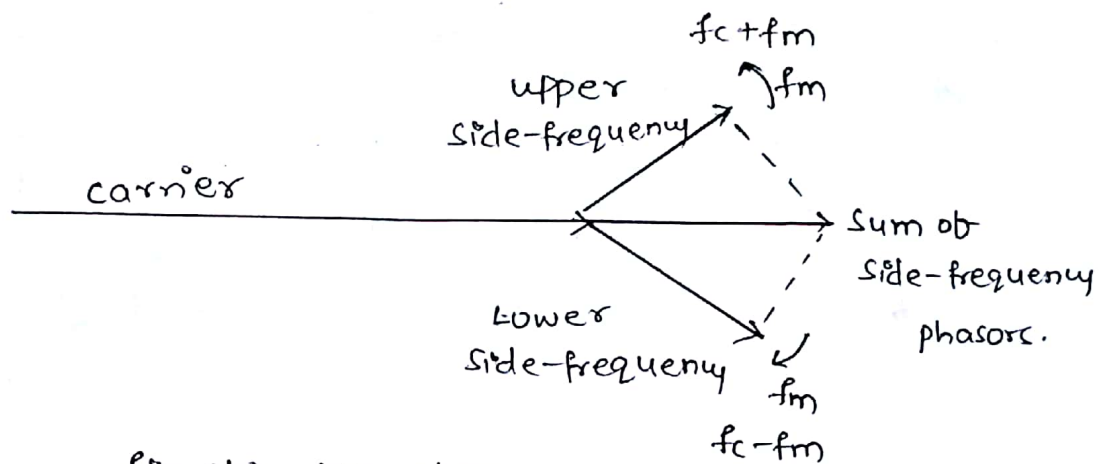
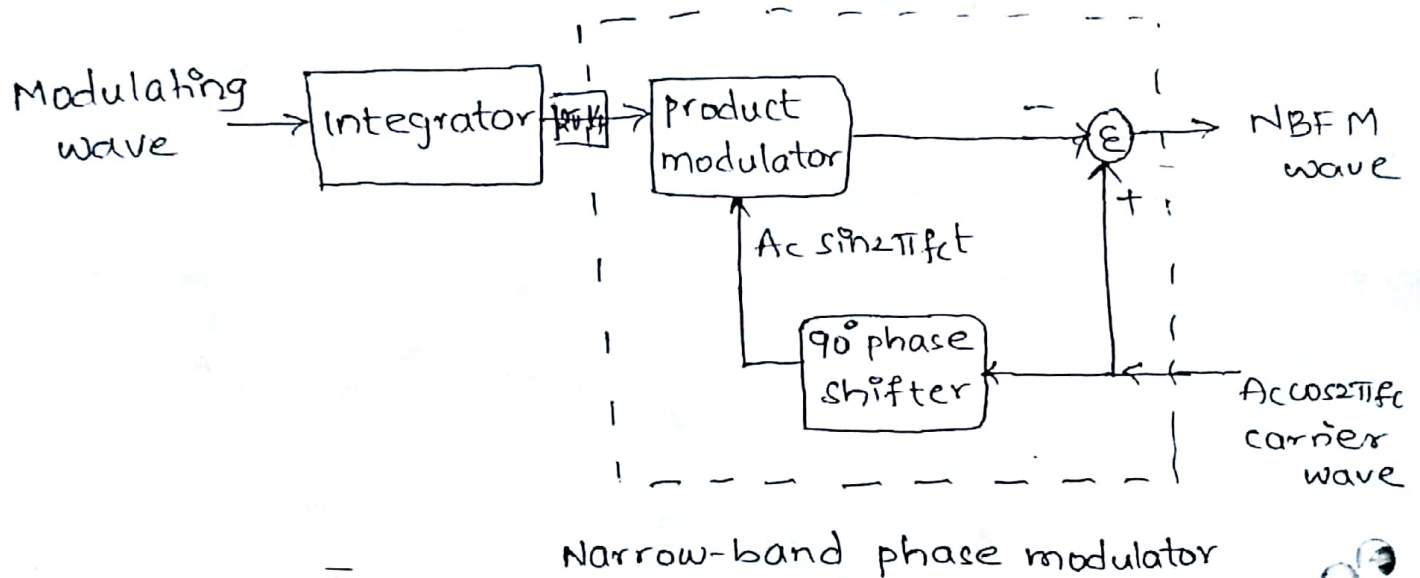


fig (b). AM phasor.

The phasor representation of NBFM is shown in fig (a). where we have used the carrier phasor as reference. we see that the resultant of the two side frequency phasors is always right angles to the carrier phasor. The effect of this is to produce a resultant phasor representing the narrowband FM wave which is approximately of the same amplitude as the carrier phasor, but out of phase with respect to it.

This phasor diagram should be contrasted with that of fig (b), which represents an AM wave. In this case we see that the resultant phasor representing the AM wave has an amplitude different from that of the carrier phasor, but always in phase with it.

eq ① defines the approximate form of a narrow-band FM wave produced by a sinusoidal modulating wave $A_m \cos 2\pi f_m t$. For this representation we obtain the modulator shown in block diagram.



This modulator involves splitting the carrier $A_c \cos 2\pi f_c t$ into two paths. one path is direct, and the other contains a 90° phase shifting network and a product modulator, which generates a DSBSC wave. A combination of these two signals produce a narrow-band FM wave with some distortion.

Narrow-band phase modulation :- (NBPM) :-

We know that the expression for ~~MBPM~~ is

$$S_{PM}(t) = A_c \cos[2\pi f_c t + k_{PM} m(t)]$$

$$= A_c [\cos 2\pi f_c t \cdot \cos[k_{PM} m(t)] - \sin 2\pi f_c t \cdot \sin[k_{PM} m(t)]]$$

let $\theta = k_{PM} m(t)$

for smaller values of θ

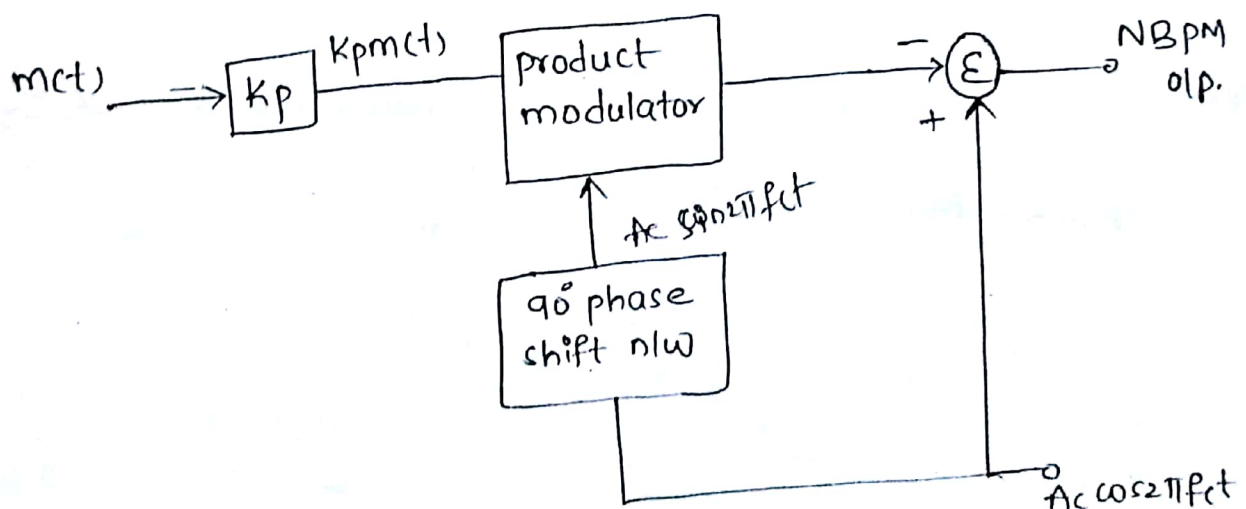
$$\sin \theta \approx \theta$$

$$\cos \theta \approx 1.$$

$$\therefore S_{PM}(t) = A_c [\cos 2\pi f_c t \cdot 1 - \sin 2\pi f_c t \cdot k_{PM} m(t)]$$

NBPM is

$$S_{PM}(t) = A_c \cos 2\pi f_c t - A_c k_{PM} m(t) \sin 2\pi f_c t$$



wide band frequency modulation:-

The wide band FM has a value of $\beta \gg 1$ hence the approximated values of $\cos(\)$, $\sin(\)$ does not hold good for wideband FM. Hence the FM wave is represented in an exponential form and finally the wide band FM can be obtained by taking the real part of exponential wave form.

For tone modulation

$$s(t) = A_c \cos[2\pi f_c t + \beta \sin 2\pi f_m t]$$

$$s(t) = A_c \operatorname{Re} \left\{ e^{j(2\pi f_c t + \beta \sin 2\pi f_m t)} \right\}$$

$$s(t) = A_c \operatorname{Re} \left[e^{j\beta \sin 2\pi f_m t} \cdot e^{j2\pi f_c t} \right]$$

$$s(t) = A_c \operatorname{Re} \left[s'(t) e^{j2\pi f_c t} \right] \quad \text{--- (1)}$$

$$\text{where } s'(t) = e^{j\beta \sin 2\pi f_m t}$$

$s'(t)$ is a periodic function, it can be expressed in exponential Fourier series as

$$s'(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi n f_m t} \quad \text{--- (2)}$$

where

$$C_n = f_m \int_{-1/2 f_m}^{1/2 f_m} s'(t) e^{-j2\pi n f_m t} dt$$

Note:-

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt$$

then

$$C_n = f_m \int_{-1/2 f_m}^{1/2 f_m} e^{j\beta \sin 2\pi f_m t} \cdot e^{-j2\pi n f_m t} dt$$

Put $2\pi f_m t = x$

$$2\pi f_m dt = dx \Rightarrow dt = \frac{dx}{2\pi f_m}$$

$$t = -1/2 f_m \Rightarrow x = -\pi$$

$$t = 1/2 f_m \Rightarrow x = \pi$$

$$\therefore C_n = f_m \int_{-\pi}^{\pi} e^{j\beta \sin x} \cdot e^{-jn x} \frac{dx}{2\pi f_m}$$

$$C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} dx$$

The value of integral $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} dx$

can be directly obtained from the table for known values of β as n is varied from $-\infty$ to ∞ . Hence this integral represents a standard "Bessel function" and it is denoted by $J_n(\beta)$.

$$\therefore C_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - nx)} dx = J_n(\beta) \quad \text{--- (3)}$$

Substitute (3) in (2) we get

$$s'(t) = \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t}$$

from eq (1)

$$s(t) = A_c \operatorname{Re} [s'(t) e^{j2\pi f_c t}]$$

$$s(t) = A_c \operatorname{Re} \left[\sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t} e^{j2\pi f_c t} \right]$$

$$\therefore s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \operatorname{Re} [e^{j2\pi (f_c + n f_m) t}]$$

∴ The expression for WBFM is

$$S(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos 2\pi(f_c + n f_m)t$$

Nature of Bessel function:-

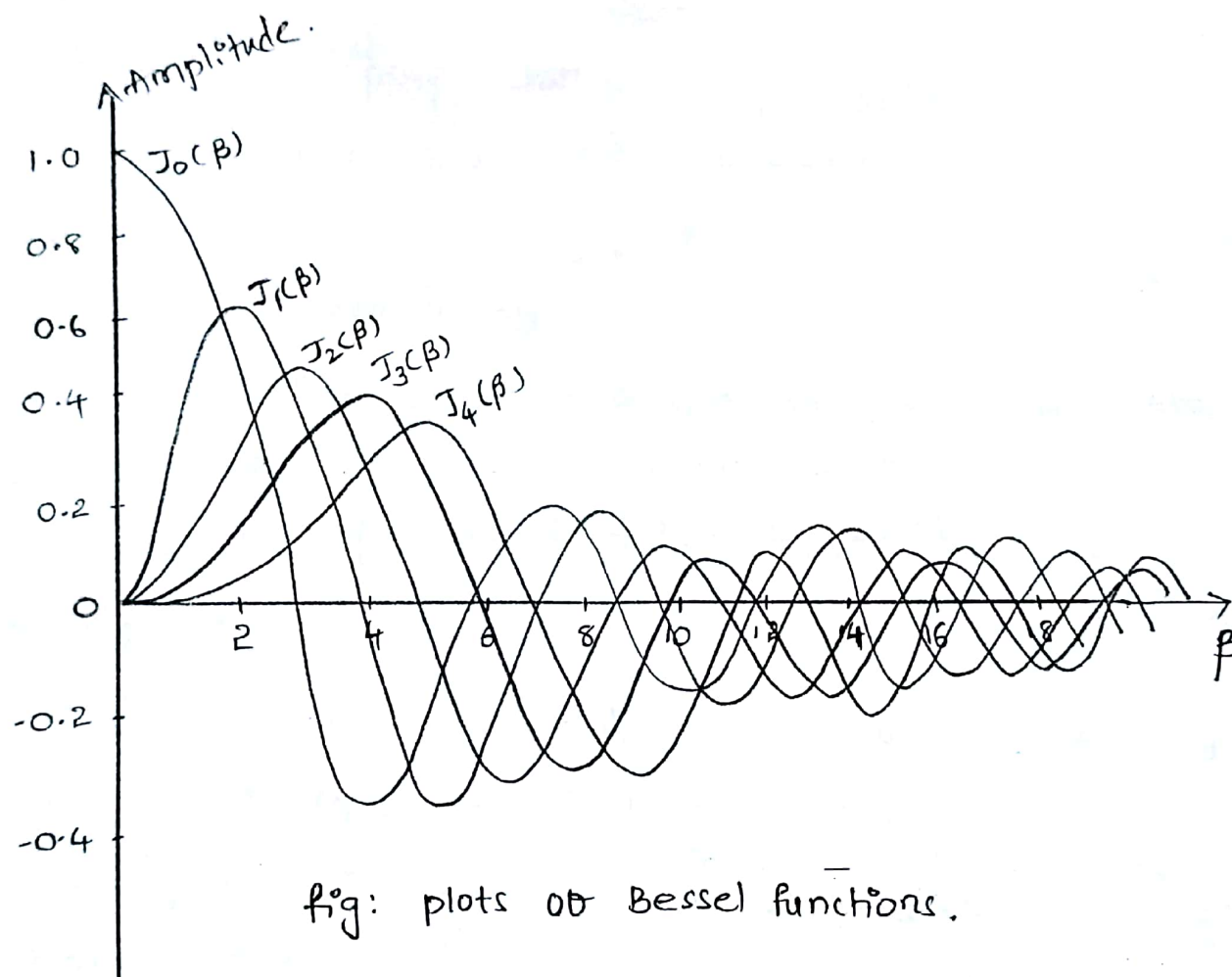


fig: plots of Bessel functions.

Characteristics of Bessel function:-

i) The Bessel coefficient $J_0(\beta)$ alone starts from a value of 1 (at $\beta = 0$).

ii) The Bessel coefficients are even functions and they satisfies.

$$|J_n(\beta)| = |J_{-n}(\beta)|$$

iii) As shown in fig the Bessel coefficients $J_n(\beta)$ passes through zero for certain values of β . This represents a temporary suppression of the FM wave-form, since the amplitude of an FM wave is governed by

$$\frac{A_c}{2} \sum_{n=-\infty}^{\infty} J_n(\beta).$$

iv) $\sum_{n=-\infty}^{\infty} J_n^2(\beta) = 1$ i.e the mean square value of all the Bessel coefficients will be equal to unity.

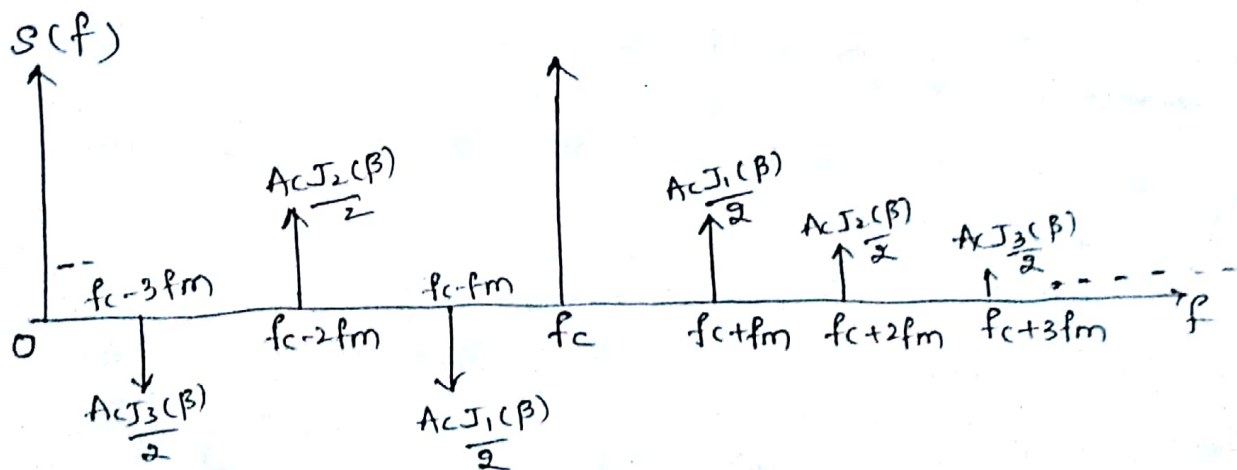
v) for n even, $J_n(\beta) = J_n(\beta)$, for n odd $J_n(\beta) = -J_n(\beta)$ i.e $J_n(\beta) = (-1)^n J_{-n}(\beta)$.

Spectrum of WBFM :-

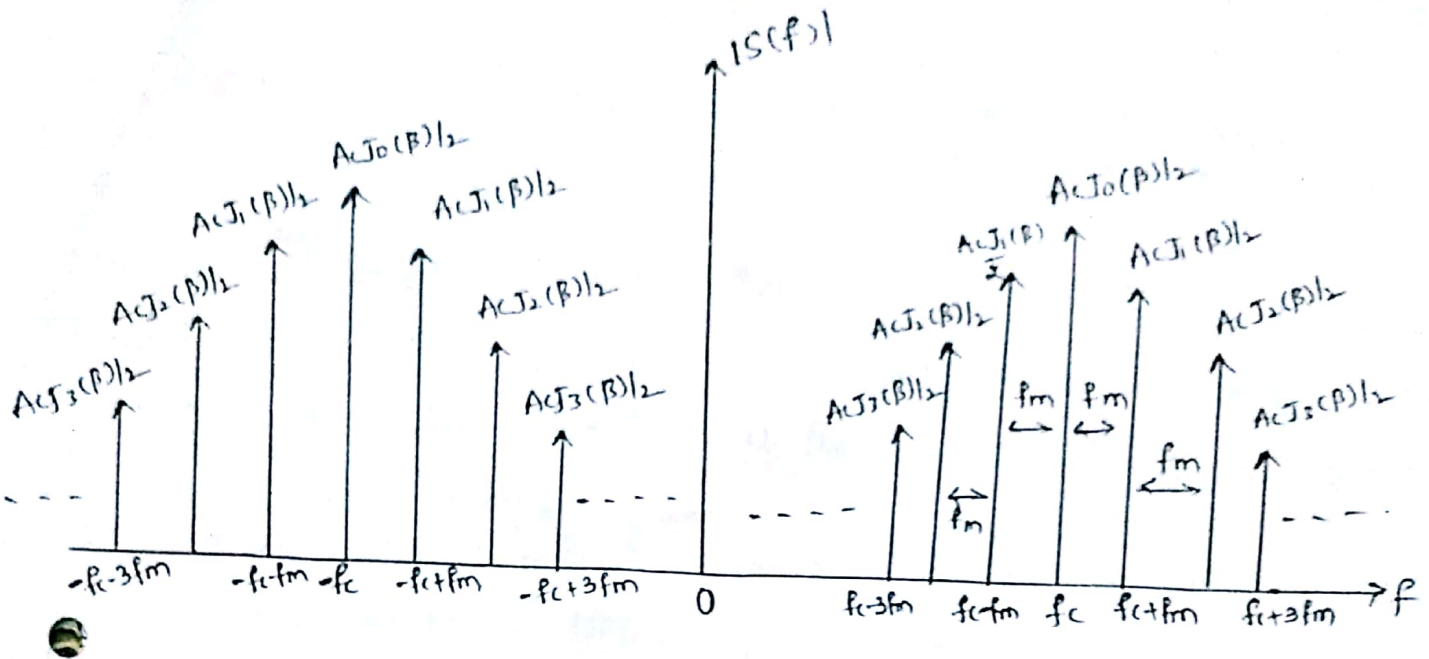
We know that the expression for WBFM is

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos[2\pi(f_c + n f_m)t]$$

$$s(t) = A_c J_0(\beta) \cos 2\pi f_c t + A_c J_1(\beta) \cos 2\pi(f_c + f_m)t + A_c J_{-1}(\beta) \cos 2\pi(f_c - f_m)t + A_c J_2(\beta) \cos 2\pi(f_c + 2f_m)t + A_c J_{-2}(\beta) \cos 2\pi(f_c - 2f_m)t + \dots$$



Let the magnitude of the spectrum should be like this



(i) From the spectrum of an FM wave it is clear that an FM signal has infinite number of frequency components spaced on either side of f_c . Hence the theoretical band width required for transmitting an FM wave is equal to infinity (∞). Thus unlike an AM waveform an FM waveform requires a channel of infinite bandwidth of receiving all the frequency components.

(ii) The spacing between the adjacent frequency components is always equal to f_m .

(iii) For large values of n say ($n > 10$) the Bessel coefficient have a very low amplitude and hence in general the number of significant components is usually restricted.

Thus even though an FM signal theoretically require ∞ bandwidth but in practice the bandwidth

is limited due to the insignificant amplitude of $J_n(\beta)$ for increasing values of n .

Calculation of practical Bandwidth using carson's rule :-

The ideal bandwidth of FM signal is infinite. In order to multiplex more no. of signals through a channel, the bandwidth of signal should be as minimum as possible.

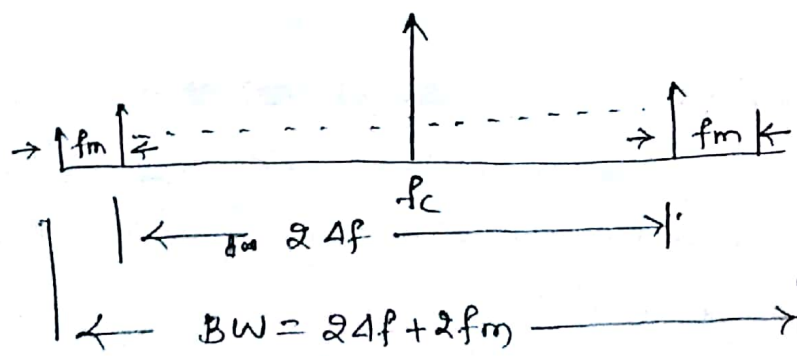
Using carson's rule it is possible to determine number of sidebands having significant magnitudes.

According to carson's rule ' $\beta+1$ ' upper and lower side bands along with carrier will have significant magnitudes and contains 99% of total power.

Using carson's rule bandwidth is given by

$$BW = 2\Delta f + 2f_m$$

where Δf is the frequency deviation and



$$BW = 2\Delta f + 2f_m$$

$$BW = 2f_m \left(1 + \frac{\Delta f}{f_m}\right)$$

$$BW = 2\left(1 + \frac{\Delta f}{f_m}\right)f_m$$

$$*** \boxed{BW = 2(1 + \beta)f_m}$$

Case i:-

We know that in NBFM $\beta \leq 1$

for practical $\beta \ll 1$.

$$BW = 2(1 + \beta)f_m$$

$$\boxed{BW \approx 2f_m}$$

Case ii:-

We know that in WBFM

$$\beta \gg 1$$

$$BW = 2f_m \cdot \beta, \text{ but } \beta = \frac{\Delta f}{f_m}$$

$$\boxed{BW \approx 2\Delta f}$$

Sr. No.	Parameter/ Characteristics	Wideband FM	Narrowband FM
1.	Modulation index	Greater than 1	Less than or slightly greater than 1
2.	Maximum deviation	75 kHz	5 kHz
3.	Range of modulating frequency	30 Hz to 15 kHz	30 Hz to 3 kHz
4.	Maximum modulation index	5 to 2500	Slightly greater than 1
5.	Bandwidth BW than of narrowband FM	Large, about 15 times higher same as that of AM	Small, approximately same as that of AM
6.	Applications	Entertainment broadcasting (can be used for high quality music transmission)	FM mobile communication like police wireless, ambulance etc. (This is used for speech transmission)
7.	Pre-emphasis and De-emphasis	required	required

Power in FM:-

We know that WBFM expression is

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta) \cos[2\pi(f_c + n f_m)t]$$

In this the amplitude of the signal is given by

$$\sum_{n=-\infty}^{\infty} A_c J_n(\beta).$$

We know that

$$\text{power } P = \frac{V_{rms}^2}{R}; \quad V_{rms} = \frac{V_m}{\sqrt{2}}$$

$\therefore$ The power in FM is

$$P = \frac{\left[A_c \sum_{n=-\infty}^{\infty} J_n(\beta) / \sqrt{2} \right]^2}{R}$$

$$P = \frac{A_c^2}{2R} \sum_{n=-\infty}^{\infty} J_n^2(\beta)$$

From the characteristic of Bessel function we know that

$$\sum_{n=-\infty}^{\infty} J_n^2(\beta) = 1.$$

$$\therefore \text{power } P = \frac{A_c^2}{2R}$$

Thus power in an FM wave is a constant and is equal to $\frac{A_c^2}{2R}$.

4-16

Unlike an AM waveform the power in FM wave is independent of the modulation index β . Hence an FM transmitter needs to transmit less amount of power compare to an AM transmitter.

Multitone FM wave :-

In practice, the modulating wave $m(t)$ is usually multitone in nature in that it consists of a group of sine-waves of different frequencies, which may be completely unrelated or harmonically related. Consider a carrier wave $A_c \cos 2\pi f_c t$ that is frequency modulated by two tones of frequencies f_1 and f_2 producing frequency deviations Δf_1 and Δf_2 respectively. The resulting FM wave can be written as

$$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int m(t) dt]$$

$$m(t) = A_1 \cos 2\pi f_1 t + A_2 \cos 2\pi f_2 t.$$

$$\therefore s(t) = A_c \cos[2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t]$$

$$s(t) = A_c \operatorname{Re} \left[e^{j(2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t)} \right]$$

$$s(t) = A_c \operatorname{Re} \left[e^{j2\pi f_c t} \cdot e^{j\beta_1 \sin 2\pi f_1 t} \cdot e^{j\beta_2 \sin 2\pi f_2 t} \right]$$

$$\text{let } s_1'(t) = e^{j\beta_1 \sin 2\pi f_1 t}$$

$$s_2'(t) = e^{j\beta_2 \sin 2\pi f_2 t}.$$

Thus expressing $s_1'(t)$, $s_2'(t)$ in exponential Fourier series we get

$$s_1'(t) = \sum_{n=-\infty}^{\infty} J_n(\beta_1) e^{jn2\pi f_1 t} \quad \text{and}$$

$$s_2'(t) = \sum_{m=-\infty}^{\infty} J_m(\beta_2) e^{jm2\pi f_2 t}$$

$$\therefore s(t) = A_c \operatorname{Re} \left[\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} J_n(\beta_1) J_m(\beta_2) e^{j2\pi(f_c + mf_2 + nf_1)t} \right]$$

$$\therefore \boxed{s(t) = A_c \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} J_n(\beta_1) J_m(\beta_2) \cos 2\pi(f_c + nf_1 + mf_2)t}$$

From the above expression it can be seen that the spectrum of FM wave produced by a modulating signal with two tones of frequencies f_1 & f_2 consists of following components.

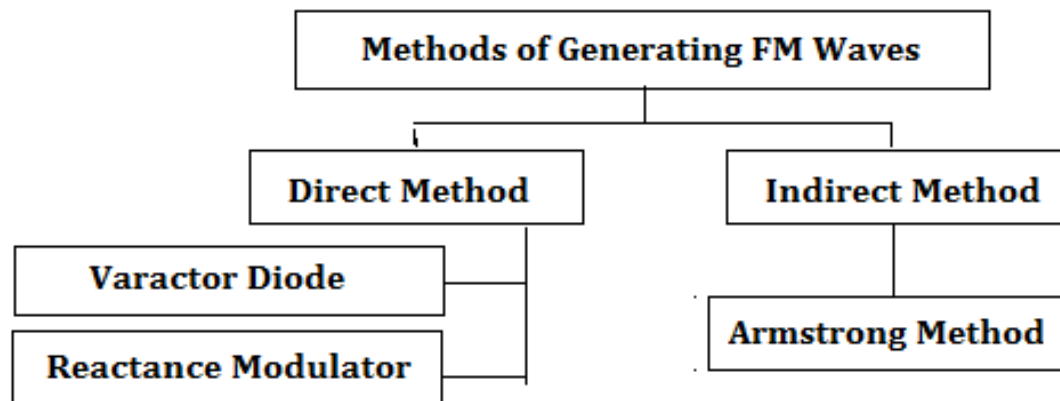
- 1) carrier components of amplitude $J_0(\beta_1)$, $J_0(\beta_2)$ of frequency f_c [$m=n=0$]
- 2) $f_c \pm nf_1$ when $n \rightarrow$ varying from $-\infty$ to ∞ & $m=0$
- 3) $f_c \pm mf_2$ when $n=0$, m varying from $-\infty$ to ∞ .
- 4) $f_c \pm nf_1 \pm mf_2$ both n & m varying.

GENERATION OF FREQUENCY MODULATION:

Generation of FM is achieved by using two methods.

1. Direct FM: In this method the carrier frequency is directly varied in accordance with the incoming message signal.

2. Indirect FM: This method was first proposed by Armstrong. In this method the modulating wave is first used to produce a narrow-band FM wave, and frequency multiplication is next used to increase the frequency deviation to the desired level.

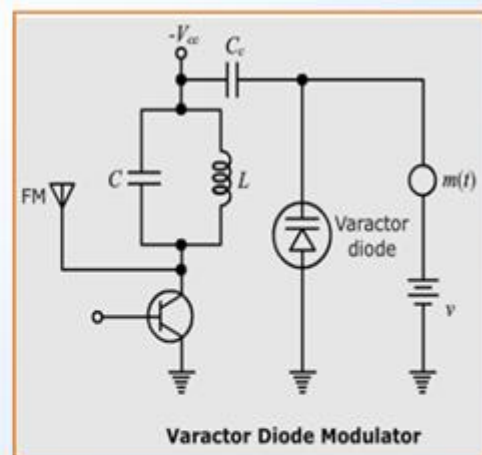


DIRECT METHOD OF FM GENERATION:

Varactor diode FM generator:

- A Varactor diode is a specially designed p-n junction diode which offers voltage variable junction capacitance. Varactor diodes are very useful as voltage variable capacitor for electronic tuning.

- With no modulating signal applied, the parallel combination of C , L and the capacitance of the Varactor diode forms a resonant carrier frequency.
- When the modulating (intelligence) signal is applied to the Varactor diode, its reverse bias is varied which causes its junction capacitance to vary in a step with the intelligence signal.



- Due to the variation of the capacitance in the tank circuit, the oscillator frequency will also subsequently vary in accordance with the intelligence signal. Thus, at the collector of the transistor, FM signal is available.

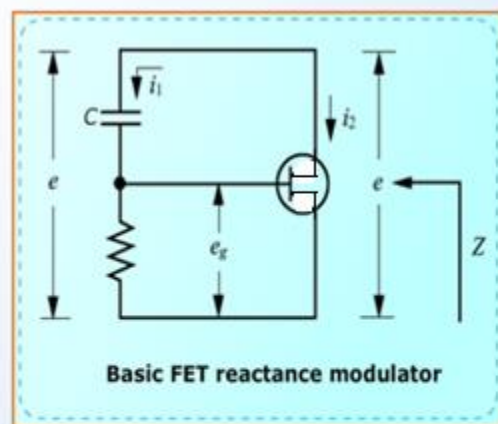
Drawbacks in Varactor diode FM:

There are certain drawbacks of implementing the varactor diode modulator.

- The FM signal is distorted when there are harmonics in the modulating signal due to non-linearity of the varactor diode.
- It is difficult to get a high order stability in carrier frequency.
- The FM station's carrier frequency variation leads to interference in the neighboring stations.

Reactance Modulator:

- The term Reactance Modulator usually refers to active device components (JFET, BJT) which offer variable reactance with the application of voltage.
- Let us consider the JFET to generate the FM signal. The JFET has the three terminal (source, drain and gate) reactance that may be connected with the tank circuit of an oscillator to be frequency modulated.



- The reactance of JFET is proportional to the trans conductance g_m of the device which in turn can be made to depend on the gate bias voltage variation.
- The basic JFET reactance modulator is shown in the figure on the right.
- From figure, the impedance Z looking back into the drain of JFET, can be written as:
$$Z = e/i_d \dots \dots \dots (1)$$
- Assuming the gate current of the JFET to be negligible, we may write:
$$e_g = i_1 R \dots \dots \dots (2)$$
- Further, the current i_1 can be written as:
$$i_1 = e/R - jX_c \dots \dots \dots (3)$$
- Where X_c is the reactance of the capacitor given by:
$$X_c = 1/2\pi fC \dots \dots \dots (4)$$
- Substituting the value of " i_1 " from equation (3) into equation (2) we get:
$$e_g = eR/R - jX_c \dots \dots \dots (5)$$
- The JFET drain circuit i_d can be written as:
$$i_d = g_m e_g \dots \dots \dots (6)$$

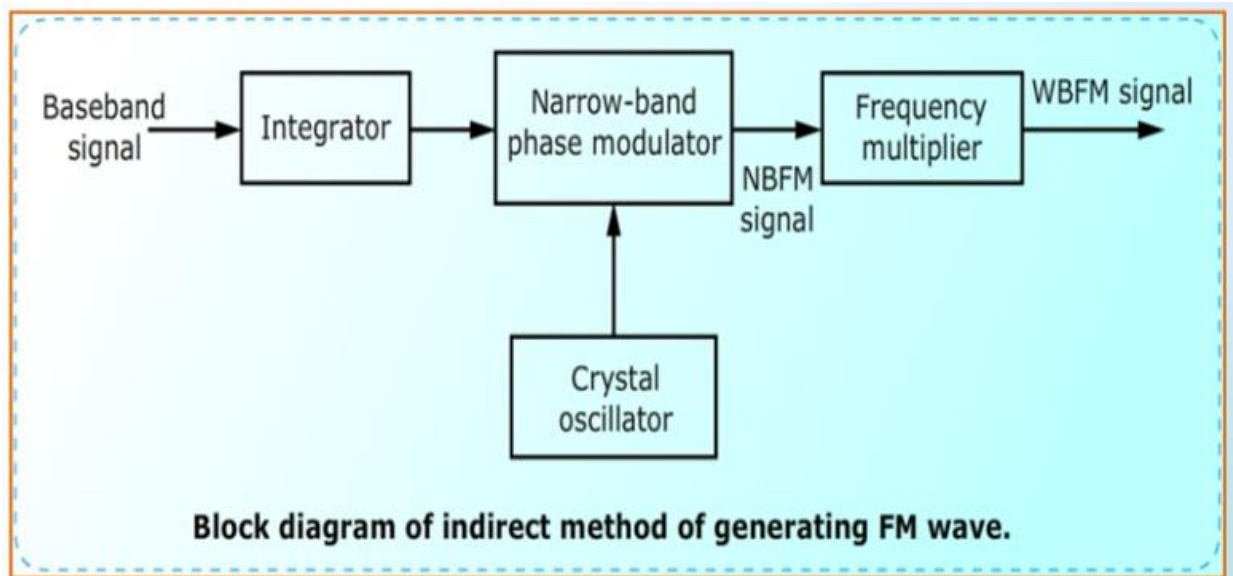
- Where g_m is the trans conductance of the device.
- Substituting the value of e_g from equation (6) into equation (5) we get:
- $i_d = g_m e_R / R - jX_c \dots \dots \dots (7)$
- The input impedance 'Z', if the JFET can be obtained from equation (1) by substituting the value of i_d from equation (7). Thus:
- $Z = 1/g_m - jX_c / g_m R \dots \dots \dots (8)$
- If the values of R and C are so chosen that $R \ll X_c$, the equation (8) reduces to:
- $Z = -j(X_c / g_m R) = 1 / (2\pi f g_m R C) = 1 / (2\pi f C_{eq}) \dots \dots \dots (9)$
- Where $C_{eq} = g_m R C \dots \dots \dots (10)$
- Note that the quantity C_{eq} has dimension of capacitance since $g_m R$ dimensionless. Thus, the impedance, Z of the JFET behaves as a capacitive reactance. If a modulating signal is applied to the JFET gate the amount of capacitance will vary in step because the JFET trans conductance, g_m varies in accordance with the applied gate voltage.
- The capacitance reactance varying with the applied voltage because of g_m variation in the JFET across the oscillator tank circuit results in FM signal.
- The major disadvantage of the reactance modulator is that they cannot be operated on crystal oscillator.
- The drawback of the direct "method is that" it uses an LC oscillator which is not stable enough for communication or broadcast purposes. Also, its circuit is complex.

INDIRECT METHOD OF FM GENERATION (ARMSTRONG METHOD):

Armstrong method of generating WBFM was first proposed by E.H. Armstrong. The following are the steps for generating WBFM waves using this method.

1. Narrowband FM is generated by integrating the baseband signal $m(t)$ and using it to phase modulate the carrier generated by a crystal oscillator.
2. In order to minimize the distortion in the phase modulation the maximum phase deviation or the index of modulation β is kept low. This results in NBFM.
3. The NBFM is converted to WBFM by using frequency multipliers. It should be noted that the carrier frequency of the resulting WBFM changes to a higher value during frequency multiplication.

- A Proper design is necessary for frequency multiplication so that desired values of deviation and the carrier frequency can be achieved simultaneously through frequency multiplication.



- Let $s_1(t)$ denote the output of the phase modulator which is the NBFM. Thus,
- $s_1(t) = A_1(\cos(2\pi f_1 t) + 2\pi k_1 \int m(t) dt) \dots \dots \dots (1)$
- Where, f_1 is the frequency of the crystal oscillator and k_1 is a constant.
- For sinusoidal modulating signal
- $s_1(t) = A_1(\cos(2\pi f_1 t) + \beta_1 \sin(2\pi f_m t)) \dots \dots \dots (2)$
- Where, β_1 is the index of modulation of NBFM, which is kept less than 0.3 radian to

keep the distortion minimum.

- The phase modulator output is next multiplied n times in frequency by using frequency multiplier, producing the desired WBFM, given by:

- $S(t) = A_c(\cos(2\pi n f_1 t + 2\pi n k_1 \int m(t) dt)) \dots \dots \dots (3)$

- Writing $n f_1 = f_c$ and $n k_1 = k_f$ for sinusoidal modulating signal, we get the desired signal as: $S(t) = A_c(\cos(2\pi f_c t + 2\pi k_f \int m(t) dt)) \dots \dots \dots (4)$

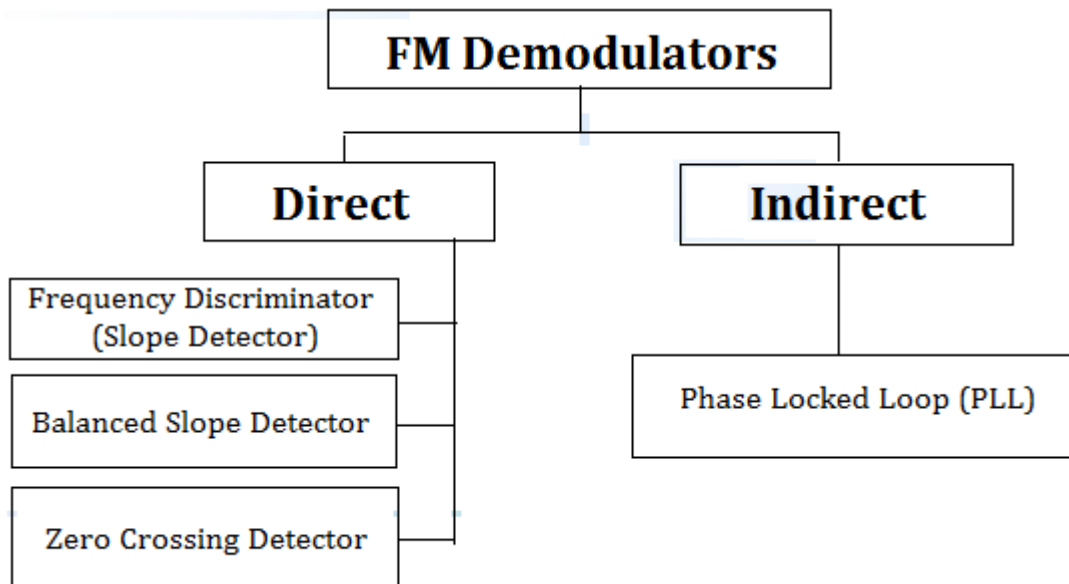
i.e. , $S(t) = A_c(\cos(2\pi f_c t) + \beta \sin(2\pi f_m t)) \dots \dots \dots (5)$

Where, $\beta = n\beta_1$

- The only drawback of the Armstrong method is that it is applicable to $\beta < 1$.

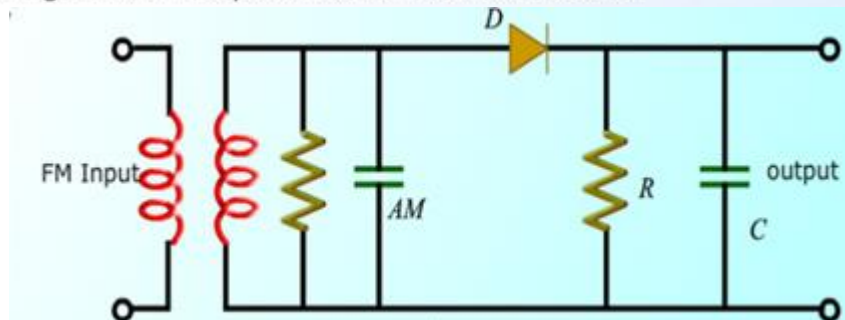
FM DEMODULATION:

- Frequency demodulation is the process to extract the original modulating signal from the frequency modulated wave. FM demodulators are classified as shown below.



Slope Detector:

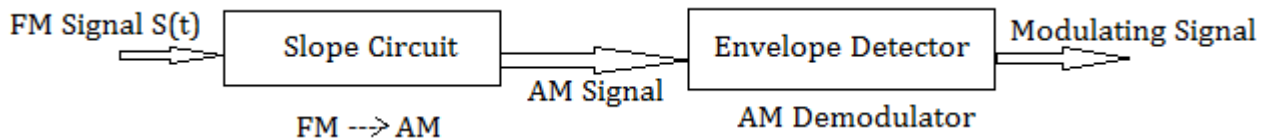
- Slope Detector is also known as frequency discriminator. It operates on the principle of frequency to voltage conversion. Slope detector is a parallel tuned circuit and consists of two units. They are Slope Circuit and Envelope Detector.
- The slope circuit converts the frequency variations in FM signal into a voltage signal. The voltage signal which is generated at the output of slope circuit is proportional to the input frequency. Examples of a slope circuit are differentiator and filters.
- The output of the slope circuit is sent to the envelope detector. The output of the envelope detector is proportional to the message signal.
- The circuit diagram of a slope detector is shown below.



Let's now look at the steps involved in the working of a slope detector.

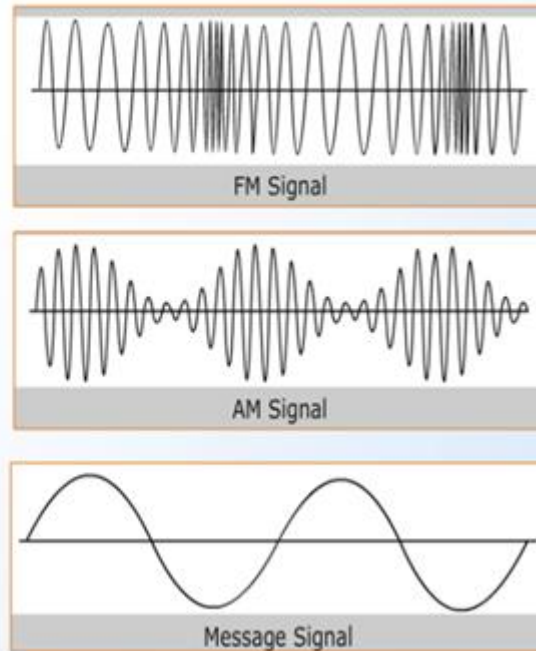
- Let $S(t)$ is a frequency modulated signal with f_c less than the resonant frequency is given to the tuned circuit (Slope Circuit) of frequency f_0 .
- As the instantaneous frequency $f_i(t)$ of the FM signal swings above or below f_c , the amplitude variation on the top of the FM signal resulting a AM wave form $s_c(t)$.
- The AM wave form is given to envelope detector to extract the message signal. The output of envelope detector is message signal $s_o(t)$.

Displayed below is the block diagram of a slope detector explaining its working.



Block diagram of Slope Detector

- The waveforms of slope detector are:



Advantages

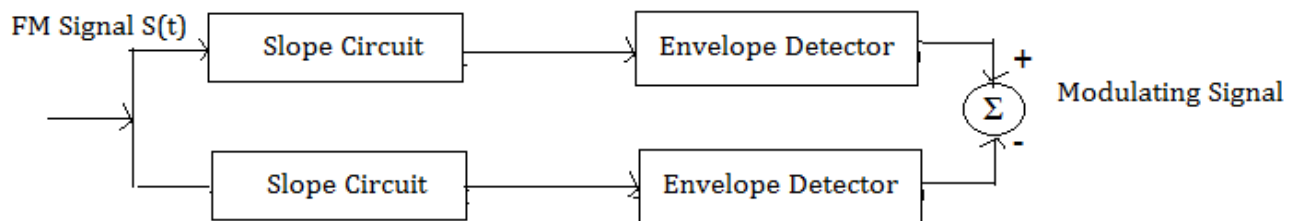
- Simple and Inexpensive

Disadvantages

- It is inefficient as it is linear in very limited frequency range.
- It reacts to all amplitude changes.
- It is relatively difficult to tune, as tuned circuit must be tuned to different frequency than carrier frequency.

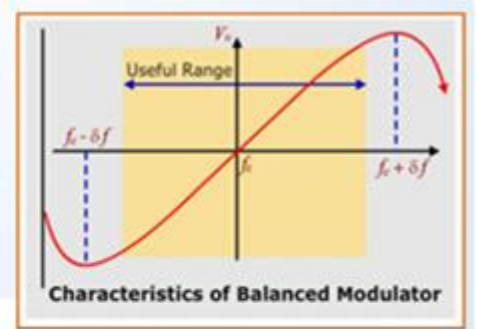
Balanced Slope Detector:

- Balanced slope detector consists of two slope circuits which are connected back to back to opposite ends of a center tapped transformer secondary.
- Due to this arrangement, the input fed to the circuits is 180 degrees out of phase. The slope circuit is connected to envelope detectors with suitable RC loads.
- Therefore, the output of the balanced slope detector is the sum of the individual output.
- Displayed below is a block diagram of a balanced slope detector.



The following steps describe the working of a balanced slope detector.

- When the input frequency = f_c , then the voltage applied to both the diodes is same. So, the output is 0.
- When input frequency = $f_c + \delta f$, the top secondary circuit is tuned above the center carrier frequency f_c by an amount δf which is the deviation in FM.
- Then, the output of diode D_1 will be large but the output of diode D_2 will be small. So the output voltage is positive.
- When the input frequency = $f_c - \delta f$, the top secondary circuit is tuned below the center carrier frequency f_c by an amount δf which is the deviation in FM. Then, the output of diode D_1 will be small but the output of diode D_2 will be large. So the output voltage is negative.



Advantages

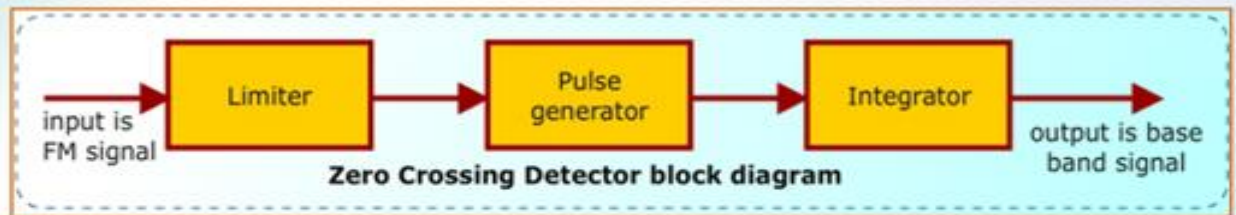
- Two slope circuits and detectors in a balanced slope detector are considered more efficient than the single slope detector.

Disadvantages

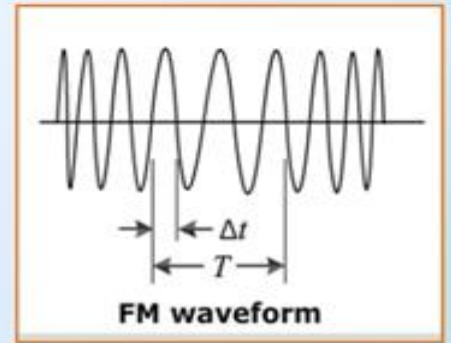
- Balanced slope detectors are more difficult to tune than the single slope detector, as there are three different frequencies to be tuned.
- Amplitude limiting still not provided in balanced slope detectors.
- Linearly although better than single slope detector, it is still not good enough.

Zero Crossing Detector:

- Zero crossing Detector is one of the direct demodulation method of FM.
- Display below is its block diagram.



- The detector exploits the property that the instantaneous frequency of an FM wave is approximately given by:
- $f_i \approx 1/2\Delta t$ (1)
- Where, Δt is the difference between adjacent zero crossings of the FM wave, as illustrated in the figure on the right.



Considering an interval T chosen in accordance with the following two conditions:

1. The interval T is small compared to the reciprocal of the message band-width W .
 2. The interval T is large compared to the reciprocal of the carrier frequency f_c of the FM wave.
- Condition 1 means that the message signal $m(t)$ is essentially constant inside the interval T .
 - Condition 2 ensures that a reasonable number of zero crossing of the FM wave occurs inside the interval T .

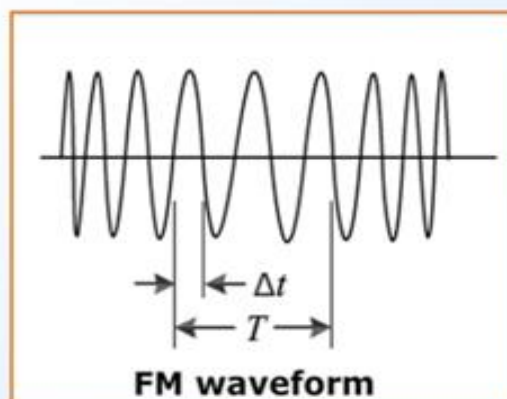
- The FM waveform shown in figure on the right illustrates these two conditions. Let n_0 denote the number of zero crossings inside the interval T .

- We may then express the time Δt between adjacent zero crossing as:

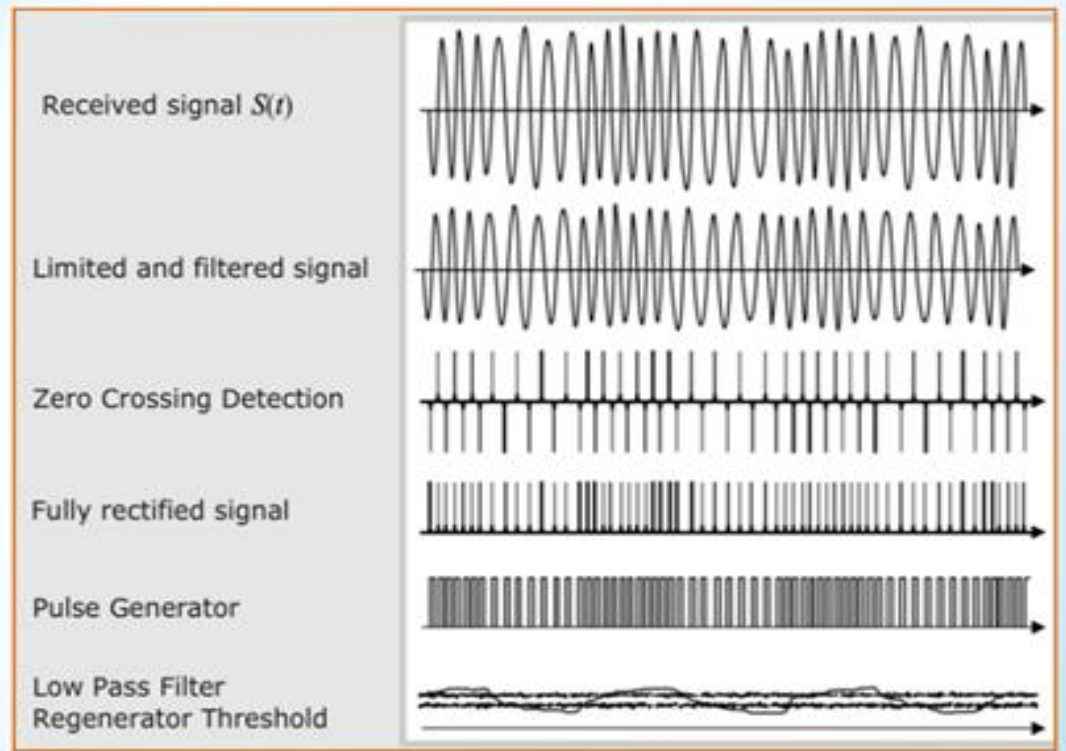
$$\Delta t = T/n_0 \text{(2)}$$

- Hence, we may rewrite equation (1)

$$f_i \approx n_0/2t \text{(3)}$$



- Zero crossing detector waveforms are shown below.



Phase locked loop:

The phase locked loop take in a signal to which it locks and can then output this signal from its own internal VCO. At first sight this may not appear particularly useful, but with a little ingenuity, it is possible to develop a large number of phase locked loop applications.

Some phase lock loop applications include:

- FM demodulation: One major phase locked loop application is that of a FM demodulator. With PLL chips now relatively cheap, this PLL applications enables high quality audio to be demodulated from an FM signal.
- AM demodulation: Phase locked loops can be used in the synchronous demodulation of amplitude modulated signals. Using this approach, the PLL locks onto the carrier so that a reference within the receiver can be generated. As this corresponds exactly to the frequency of the carrier, it can be mixer with the incoming signal to synchronous demodulate the AM.
- Indirect frequency synthesizers: Use within a frequency synthesizer is one of the most important phase locked loop applications. Although direct digital synthesis is also used, indirect frequency synthesis forms one of the major phase locked loop applications.
- Signal recovery: The fact that the phase locked loop is able to lock to a signal enables it to provide a clean signal, and remember the signal frequency if there is a short interruption. This phase locked loop application is used in a number of areas where signals may be interrupted for short periods of time, for example when using pulsed transmissions.

- **Timing distribution:** Another phase locked loop application is in the distribution of precisely timed clock pulses in digital logic circuits and systems, for example within a microprocessor system.

Phase locked loop basic concepts - phase

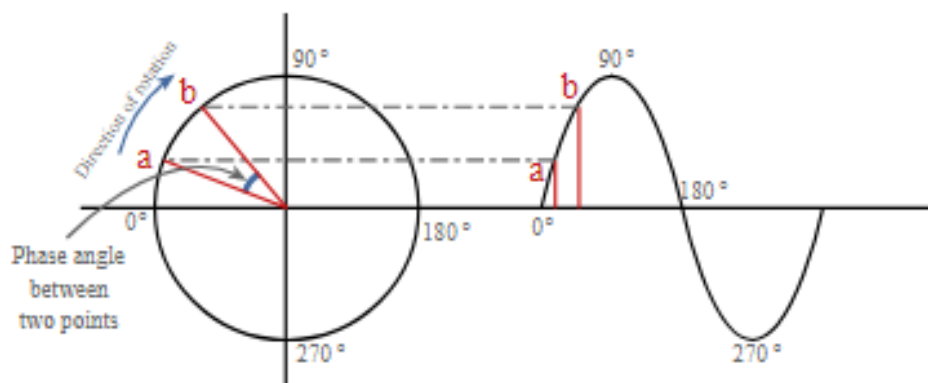
The key to the operation of a phase locked loop, PLL, is the phase difference between two signals, and the ability to detect it. The information about the error in phase or the phase difference between the two signals is then used to control the frequency of the loop.

To understand more about the concept of phase and phase difference, it is possible to visualise two waveforms, normally seen as sine waves, as they might appear on an oscilloscope. If the trigger is fired at the same time for both signals they will appear at different points on the screen.

The linear plot can also be represented in the form of a circle. The beginning of the cycle can be represented as a particular point on the circle and as time progresses the point on the waveform moves around the circle. Thus a complete cycle is equivalent to 360° or 2π radians. The instantaneous position on the circle represents the phase at that given moment relative to the beginning of the cycle.

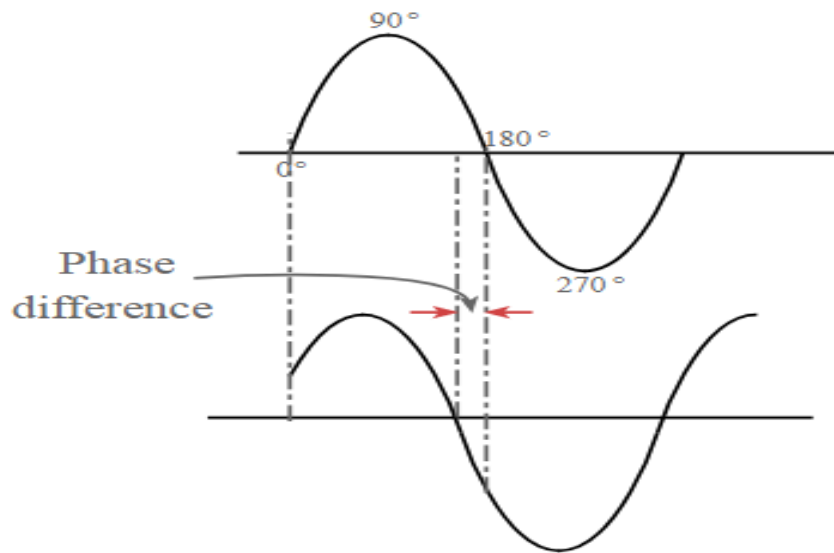
The concept of phase difference takes this concept a little further. Although the two signals we looked at before have the same frequency, the peaks and troughs do not occur in the same place.

There is said to be a phase difference between the two signals. This phase difference is measured as the angle between them. It can be seen that it is the angle between the same point on the two waveforms. In this case a zero crossing point has been taken, but any point will suffice provided that it is the same on both



Phase angle of points on a sine wave

This phase difference can also be represented on a circle because the two waveforms will be at different points on the cycle as a result of their phase difference. The phase difference measured as an angle: it is the angle between the two lines from the centre of the circle to the point where the waveform is represented.



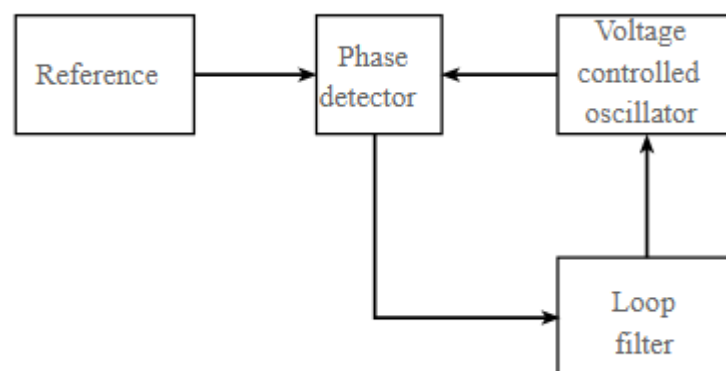
Phase difference between signals

When there two signals have different frequencies it is found that the phase difference between the two signals is always varying. The reason for this is that the time for each cycle is different and accordingly they are moving around the circle at different rates.

It can be inferred from this that the definition of two signals having exactly the same frequency is that the phase difference between them is constant. There may be a phase difference between the two signals. This only means that they do not reach the same point on the waveform at the same time. If the phase difference is fixed it means that one is lagging behind or leading the other signal by the same amount, i.e. they are on the same frequency.

Phase locked loop basics

A phase locked loop, PLL, is basically of form of servo loop. Although a PLL performs its actions on a radio frequency signal, all the basic criteria for loop stability and other parameters are the same. In this way the same theory can be applied to a phase locked loop as is applied to servo loops.



Basic phase locked loop basic diagram

A basic phase locked loop, PLL, consists of three basic elements:

- Phase comparator / detector: As the name implies, this circuit block within the PLL compares the phase of two signals and generates a voltage according to the phase difference between the two signals.

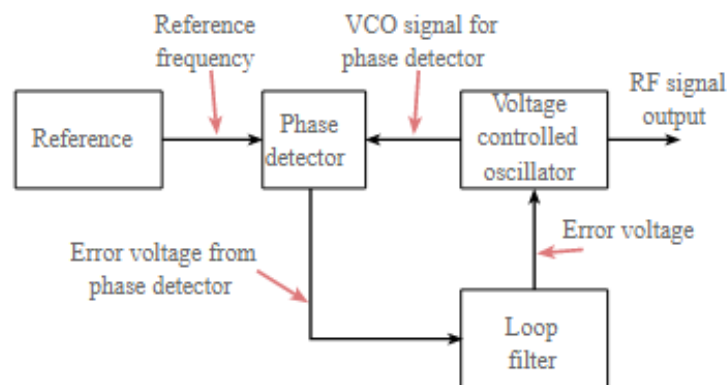
- **Voltage controlled oscillator(VCO):** The voltage controlled oscillator is the circuit block that generates the radio frequency signal that is normally considered as the output of the loop. Its frequency can be controlled over the operational frequency band required for the loop.
- **Loop filter:** This filter is used to filter the output from the phase comparator in the phase locked loop. It is used to remove any components of the signals of which the phase is being compared from the VCO line, i.e. the reference and VCO input.

Phase locked loop operation

The basic concept of the operation of the PLL is relatively simple, although the mathematical analysis and many elements of its operation are quite complicated

The diagram for a basic phase locked loop shows the three main element of the PLL: phase detector, voltage controlled oscillator and the loop filter.

In the basic PLL, reference signal and the signal from the voltage controlled oscillator are connected to the two input ports of the phase detector. The output from the phase detector is passed to the loop filter and then filtered signal is applied to the voltage controlled oscillator.



Phase locked loop diagram showing voltages

The Voltage Controlled Oscillator, VCO, within the PLL produces a signal which enters the phase detector. Here the phase of the signals from the VCO and the incoming reference signal are compared and a resulting difference or error voltage is produced. This corresponds to the phase difference between the two signals.

The error signal from the phase detector passes through a low pass filter which governs many of the properties of the loop and removes any high frequency elements on the signal. Once through the filter the error signal is applied to the control terminal of the VCO as its tuning voltage. The sense of any change in this voltage is such that it tries to reduce the phase difference and hence the frequency between the two signals. Initially the loop will be out of lock, and the error voltage will pull the frequency of the VCO towards that of the reference, until it cannot reduce the error any further and the loop is locked.

When the PLL, phase locked loop, is in lock a steady state error voltage is produced. By using an amplifier between the phase detector and the VCO, the actual error between the signals can be reduced to very small levels. However some voltage must always be present at the control terminal of the VCO as this is what puts onto the correct frequency.

The fact that a steady error voltage is present means that the phase difference between the reference signal and the VCO is not changing. As the phase between these two signals is not changing means that the two signals are on exactly the same frequency.

The phase locked loop, PLL is a very useful building block, particularly for radio frequency applications. The PLL forms the basis of a number of RF systems including the indirect frequency synthesizer, a form of FM demodulator and it enables the recovery of a stable continuous carrier from a pulse waveform. In this way, the phase locked loop, PLL is an essential RF building tool.

Comparison of FM and AM System:-

FM	AM
1. Amplitude of FM wave is constant. It is independent of the modulation index.	1. Amplitude of AM wave will change with the modulating voltage.
2. The transmitted power remains constant. It is independent of modulation index. $P_t = A_c^2/2R$	2. Transmitted power is dependent on the modulation index. $P_t = P_c [1 + M^2/2]$
3. All the transmitted power is useful.	3. carrier power and one sideband powers are useless.
4. FM receivers are immune to noise.	4. AM receivers are not immune to noise.
5. It is possible to decrease noise further by increasing deviation.	5. This feature is absent in AM
6. Bandwidth = $2(\Delta f + f_m)$. The bandwidth depends on β .	6. $BW = 2f_m$. It is independent of modulation index.
7. BW is large. Hence wide channel is required (200 KHz standard BW).	7. BW is much less than FM (channel requires 10 KHz standard).
8. Space wave is used for propagation. So radius of transmission is limited to line of sight.	8. Ground wave and sky wave propagation is used. Therefore larger area is covered than FM.

9. It is possible to operate several transmitter on same frequency ex: 103.5 MHz

10. FM transmission and reception equipment are more complex.

11. The number of sidebands having significant amplitudes depends on modulation index β .

12. The information is contained in the frequency variation of the carrier.

9. Not possible to operate more channels on the same frequency.

10. AM equipments are less complex.

11. Number of sidebands in AM will be constant and equal to 2.

12. The information is contained in the amplitude variation of the carrier.

UNIT-3

Problems and solutions

EXAMPLE-1: A music signal with frequency components from 50 Hz to 21000 Hz is Frequency modulated. If the maximum allowed frequency deviation is 50 kHz.

(i) What is the modulation index?

(ii) What is the signal bandwidth using Carson's rule?

Solution:

(i) Modulation index = frequency deviation/modulating frequency
 $= 50 \text{ kHz} / 21 \text{ kHz} = 2.38$

(ii) Bandwidth = $2 * (\text{modulating frequency} + \text{frequency deviation})$
 $= 2 * (21 \text{ kHz} + 50 \text{ kHz}) = 142 \text{ kHz}$

EXAMPLE-2: A 200 MHz carrier is frequency modulated by a 10 V peak-to-peak signal of 10 kHz. The instantaneous carrier frequency varies between 199.90 and 200.10 MHz.

Calculate (i) The modulator sensitivity.

(ii) The modulation index.

(iii) The signal bandwidth using Carson's rule.

Solution:

(i) $2 * \text{Frequency deviation} = 200.10 - 199.9 \text{ MHz} = 200 \text{ kHz}.$

Frequency deviation = 100 kHz

This is caused by a 10 V signal. Therefore sensitivity is $100 / 10 \text{ kHz/V} = 10 \text{ kHz/V}$

(ii) The modulation index is $200 \text{ kHz} / 10 \text{ kHz} = 20.$

(iii) $BW = 2 (\text{frequency of modulating signal} + \text{frequency deviation})$
 $= 2 (f_m + k * f_d)$
 $= 2 (1 + b) * f_m$

Using any of the above, $BW = 420 \text{ kHz}$

EXAMPLE-3: A speech signal with frequency components from 300 Hz to 4 kHz is Frequency modulated. If the maximum allowed frequency deviation is 30 kHz

(i) What is the modulation index?

(ii) What is the signal bandwidth using Carson's rule?

Solution:

(i) Modulation index = frequency deviation/modulating frequency = $30 \text{ kHz} / 4 \text{ kHz} = 7.5$

(ii) Bandwidth is $2 (\text{modulating frequency} + \text{frequency deviation}) = 2 (30 \text{ kHz} + 4 \text{ kHz}) = 68 \text{ kHz}.$

EXAMPLE-4: Calculate the dissipation in power across 20Ω resistor for the FM signal $v(t) = 20 \cos(6600t + 10 \sin 2100t)$.

Solution:

A standard FM signal is represented by

$$v(t) = A_c \cos(2\pi f_c t + k_f \sin 2\pi f_m t)$$

A_c = carrier amplitude

f_c = carrier frequency

k_f = modulation index

f_m = modulating frequency

k_f = frequency deviation/modulating frequency

the power dissipated across 20Ω resistor = $V_{rms}^2 / R = (20/\sqrt{2})^2 / R = 5W.$

EXAMPLE-5: A 100MHz carrier is frequency modulated by 10 KHz wave. For a frequency deviation of 50 KHz, calculate the modulation index of the FM signal.

Solution:

Carrier frequency $f_c = 100\text{MHz}$

Modulating frequency $f_m = 10\text{ KHz}$

Frequency deviation $\Delta f = 500\text{ KHz}$

Modulation index of FM signal, $m_f = \Delta f/f_m = 500 \times 10^3 / 10 \times 10^3 = 50$.

EXAMPLE-6: Determine the Bandwidth of a FM wave when the maximum deviation allowed is 75KHz and the modulating signal has a frequency of 10KHz.

Solution:

Modulating frequency $f_m = 10\text{ KHz}$

Frequency deviation $\Delta f = 75\text{ KHz}$

According to Carson's rule, $BW = 2(\Delta f + f_m) = 2(75 + 10) = 170\text{ KHz}$.

EXAMPLE-7: A 100MHz carrier is frequency modulated by 5 KHz wave. For a frequency deviation of 100 KHz, calculate the carrier swing of the FM signal.

Solution:

Carrier frequency $f_c = 100\text{MHz}$

Modulating frequency $f_m = 5\text{ KHz}$

Frequency deviation $\Delta f = 100\text{ KHz}$

Carrier swing of the FM signal $= 2 \times \Delta f = 2 \times 100 = 200\text{ KHz}$.

EXAMPLE-8: For a FM signal $v(t) = 20 \cos (10 \times 10^8 t + 30 \sin 3000t)$, calculate the power dissipated by the FM wave in a 20Ω resistor.

Solution:

Standard expression for FM signal is given by

$$v(t) = A \cos (\Omega c t + m_f \sin \Omega m t)$$

Comparing with the given equation,

$$A = 20$$

The dissipated power is given by $P = V_{\text{rms}}^2 / R$

$$= (20/\sqrt{2})^2 / 20$$

$$= 10\text{ Watts}$$

$$\text{therefore, } 1.411 = \frac{f_{si}}{30\text{ MHz}} - \frac{30\text{ MHz}}{f_{si}}$$

But from equation (ii), we have,

$$1.411 = \frac{1930}{1000} - \frac{1000}{1930}$$

Hence, equating this expression with equation (v), we obtain

$$\frac{1930}{1000} = \frac{f_{si}}{30\text{ MHz}}$$

$$\text{or } 1.9 \times 30\text{ MHz} = f_{si}$$

$$\text{New } f_{si} = 57\text{ MHz}$$

$$\text{But } f_{si} = f_{ci} + 2\text{IF}$$

$$\text{Hence, } \text{IF} = \frac{f_{si} - f_{ci}}{2} = \frac{57 - 30}{2}$$

$$\text{or } \text{IF} = 13.5\text{ MHz}$$